LMT Haptic Texture Database

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LMT Haptic Texture Database

Chair of Media Technology
Technical University of Munich

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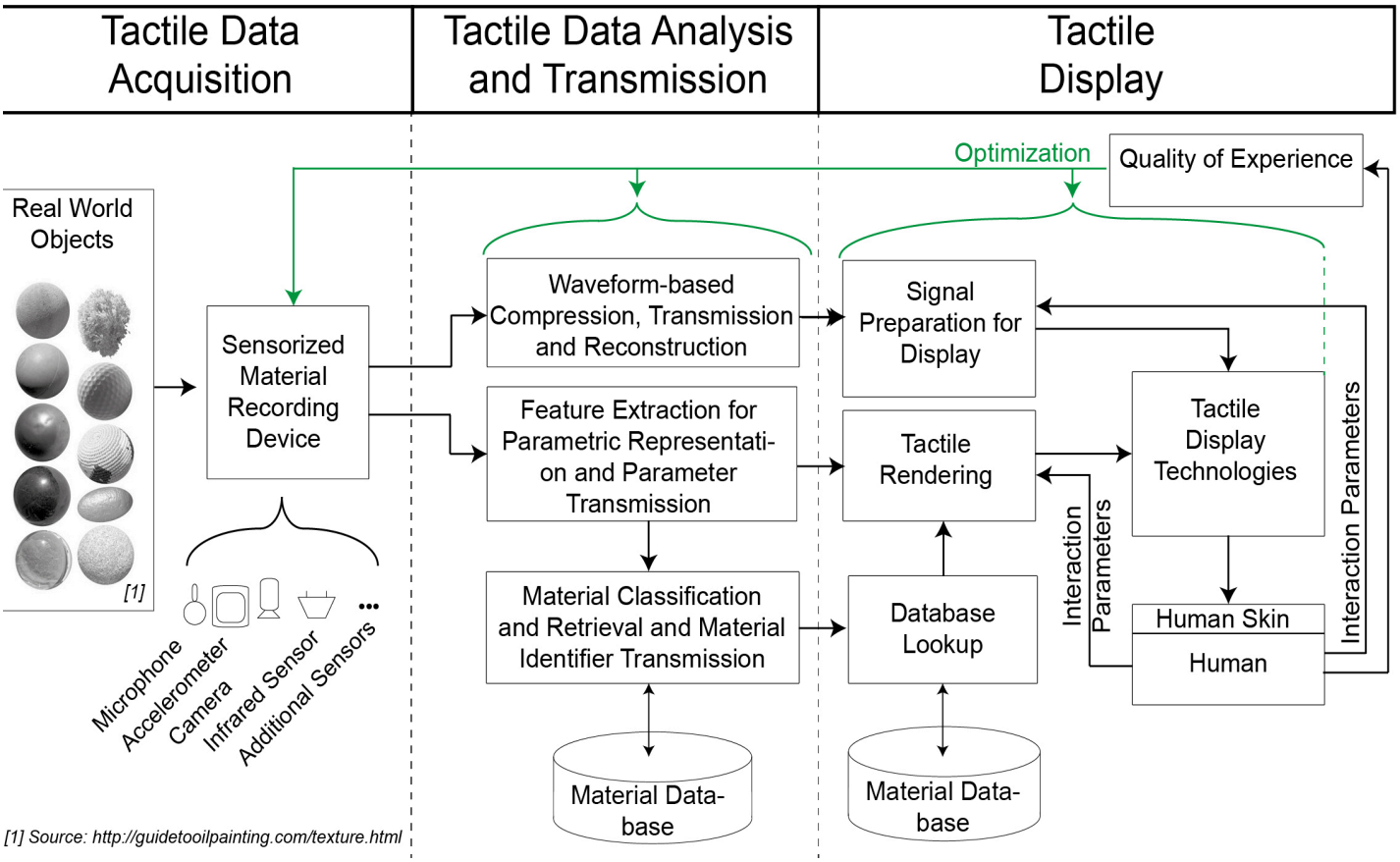
)verview of Haptic Data Acquisition, Parametrization, and Display



[1] Source: http://guidetooilpainting.com/texture.html

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:ollection of Building Material Haptic Database

'e provide a large collection of materials based on a geological, chemical and biological naming convention.

:1 Wood

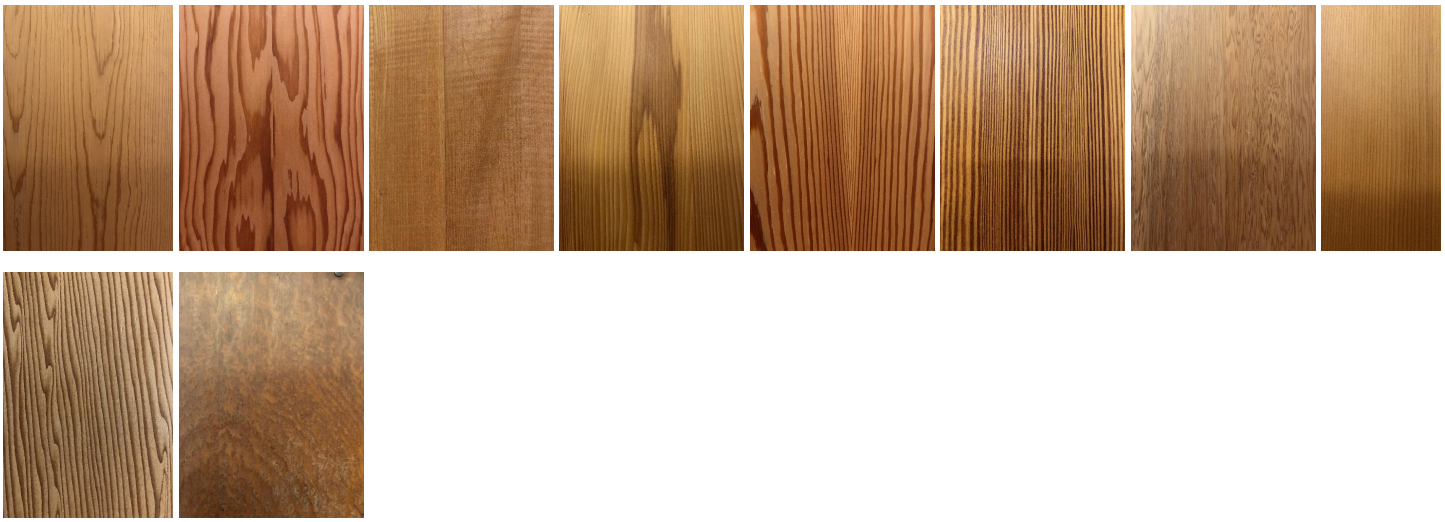
Processing Shape P

P_1 veneered, tangential section
P_2 veneered, cross section
P_3 veneered, sliced/peeled
P_4 natural, tangential section
P_5 natural, cross section
P_6 natural, sliced/peeled

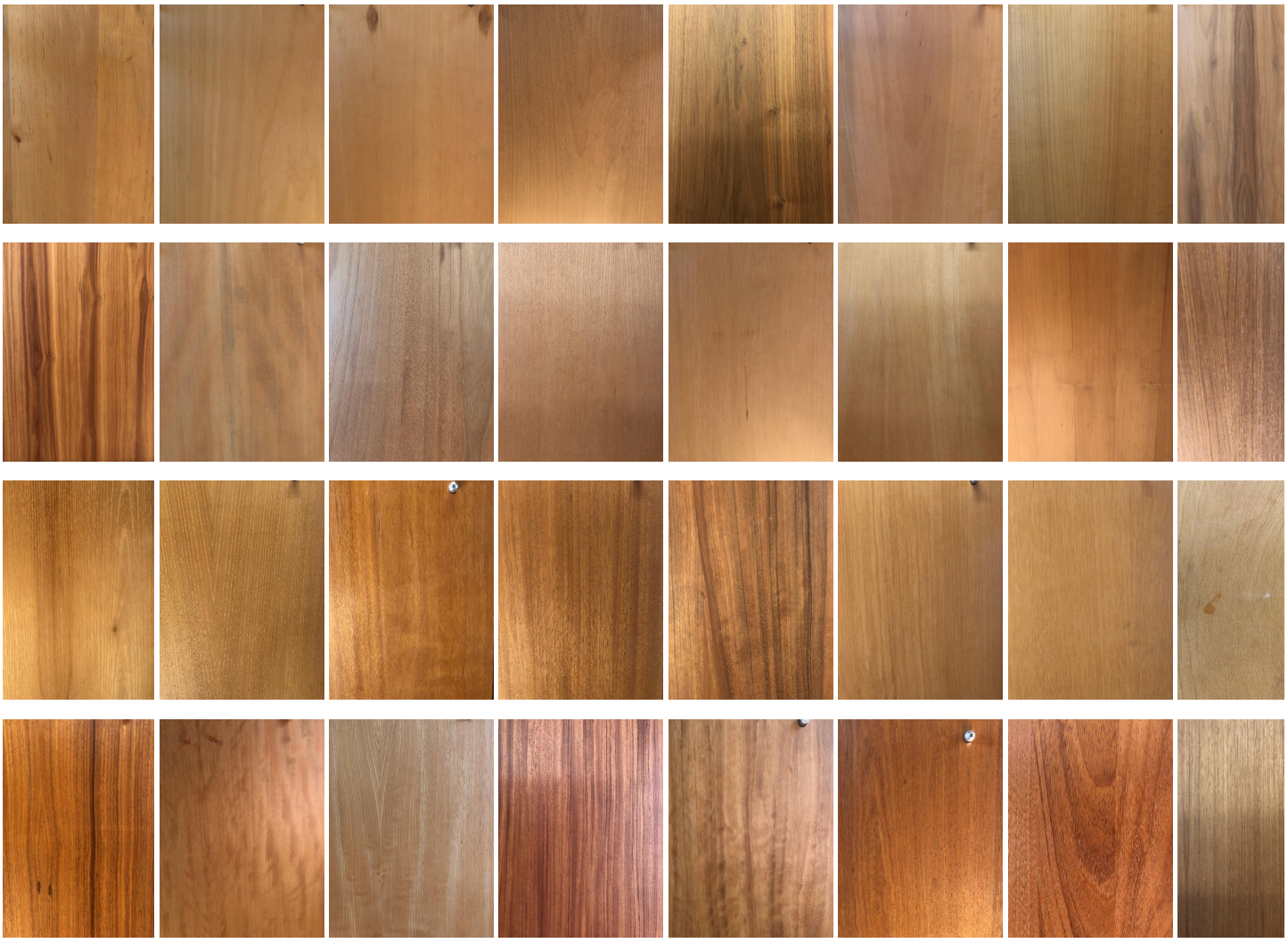
Moisture Content

X_1 dry
X_y percentage of water in y %

1 Softwood



2 Hardwood



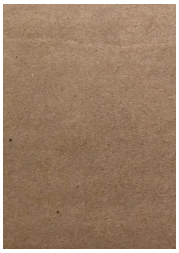
Processing Shape P	Composition
P_1 pressed	X_1 dry
P_2 natural	X_y percentage of cork in y %

3 Barks

Processing Shape P	Color
P_1 pressed	X_1 unprocessed brown
P_2 refined	X_2 bleached
P_3 crumpled	X_3 single-color painted
P_4 corrugated	X_4 multi-color painted

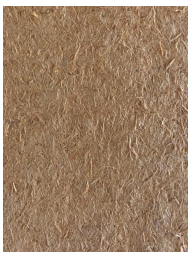
P_5 creped paper
P_6 recycled

4 Celluloids



Processing Shape P	Composition
P_1 pressed	X_1 pure
P_2 laminated	X_y percentage of main wooden component in y %

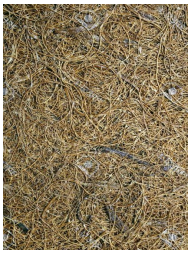
5 Crafted Wood



2 Biodegradable

Processing Shape P	Composition
P_1 molded	X_1 pure
P_2 woven	X_y percentage of main fibers in y %

1 Natural Fibers



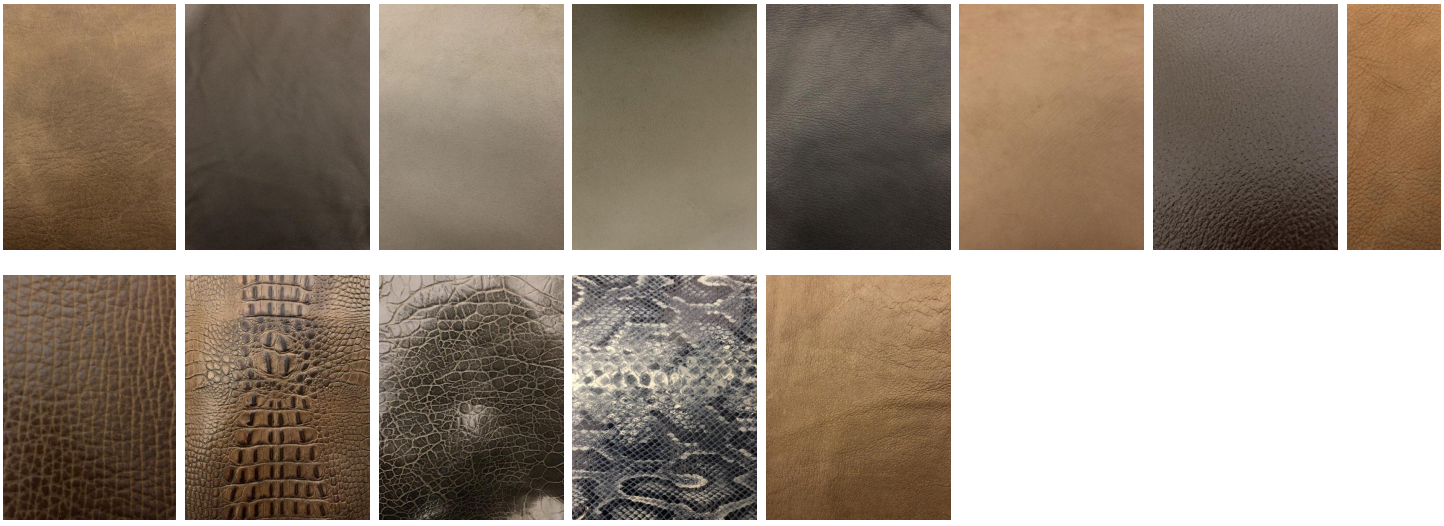
Processing Shape P	Composition
P_1 woven	X_1 pure
P_2 woven (flaked)	X_y percentage of main fur in y %
P_3 natural	

2 Fur

Side P
P_1 front side
P_2 back side

Originality X
X_1 original
X_2 artificial

3 Animal Skin
nd Leather

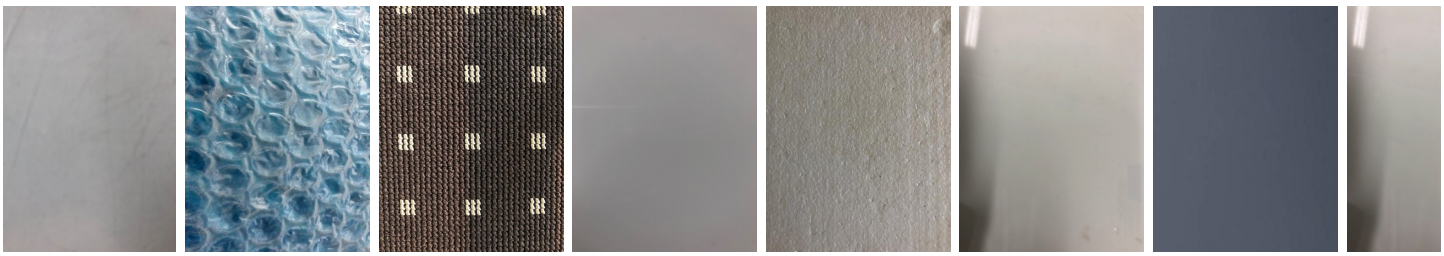


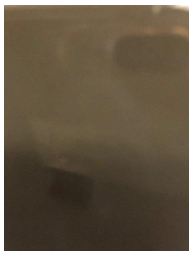
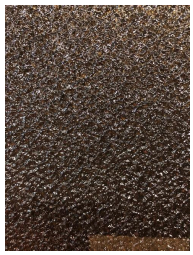
3 Polymers

Processing Shape P
P_1 injection-molded and flat
P_2 injection-molded and processed
P_3 extruded
P_4 blown film-extruded
P_5 woven
P_6 embossed
P_7 mesh (> 5mm)
P_8 mesh (< 5mm)

Composition
X_1 pure
X_y percentage of main polymer in y %

1 Thermoplasts



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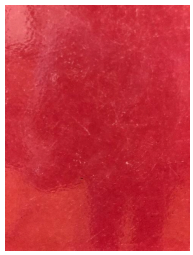
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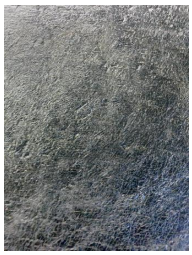
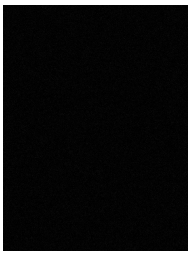


Processing Shape P
P_1 pressed
P_2 perforated
P_3 moulded

Composition
X_1 pure
X_y percentage of main polymer in y %

2 Thermosets

Not recorded yet

Not recorded yet

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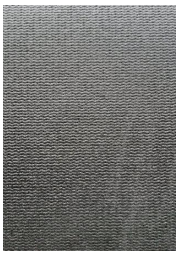
Processing Shape P
P_1 injection-molded
P_2 foamed
P_3 perforated

Composition
X_1 pure
X_y percentage of main polymer in y %

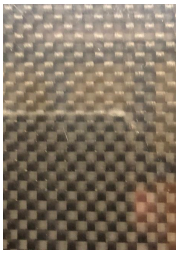
3 Elastomers



Processing Shape P
P_1 extruded

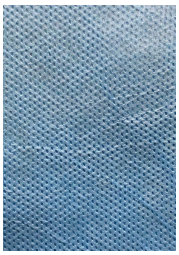
Composition
X_1 pure
X_y percentage of main carbon component in y %

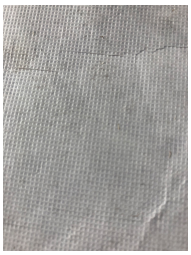
4 Carbon Fibers



Processing Shape P	Composition
P_1 woven	X_1 pure
P_2 mesh < 5mm	X_y percentage of main component in y %
P_3 mesh > 5mm	

:5 Polymer :omposites





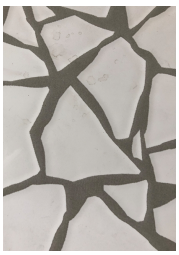
:4 Ceramics

Processing Shape P	Composition
P_1 sand-beamed or not glazed	X_1 pure
P_2 polished, glazed	X_y percentage of main component in y %

:1 Earthenware

Not recorded yet
Not recorded yet
Not recorded yet

:2 Stoneware



:3 Hightech :eramics

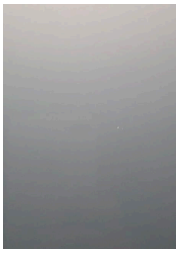
Not recorded yet

:5 Glasses

Processing Shape P	Opacity X
P_1 floated	X_1 transparent
P_2 spun	X_2 nontransparent
P_3 foamed	X_y percentage of opacity in y %

P_4 flung
P_5 corrugated
P_6 cut
P_7 ornaments

61 Crafted Glasses



62 Natural Glasses

66 Stones

Processing Shape P
P_1 stacked
P_2 polished
P_3 pointed
P_4 sanded
P_5 embossed

Composition X
X_1 pure
X_y percentage of main stone component in y %

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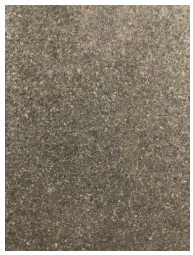
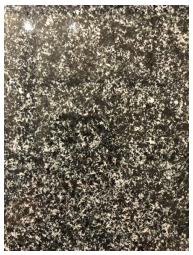
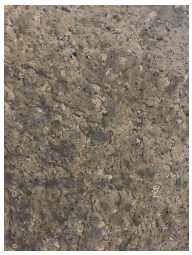
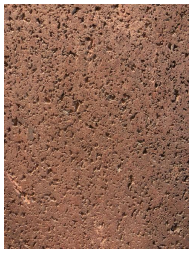



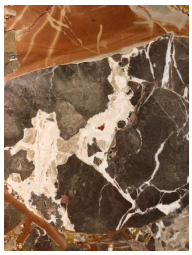


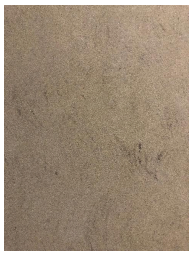


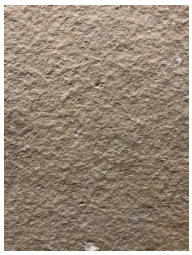
Processing Shape P
P_1 embossed
P_2 polished
P_3 sand-beamed
P_4 split
P_5 sanded

Composition X
X_1
X_y percentage of main stone component in y %

2 Sediments



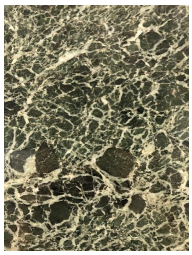


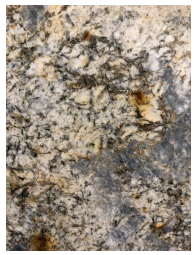


Processing Shape P
P_1 flamed
P_2 sand-beamed
P_3 polished
P_4 pointed

Composition X
X_1
X_y percentage of main stone component in y %

3 Metamorphs





Processing Shape P
P_1 beamed
P_2 pointed
P_3 embossed
P_4 axed
P_5 charringed
P_6 corrugated

Cement Standardization Specification X
X_1 CEM I Portland Cement
X_2 CEM II Portland Cement Blends
X_3 CEM III Slag-lime cements
X_4 CEM IV Pozzolan Cement
X_5 CEM V Composite Cement

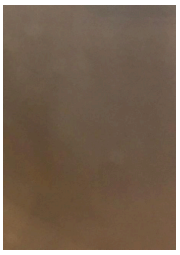
4 Cement

5 Minerals

7 Metals

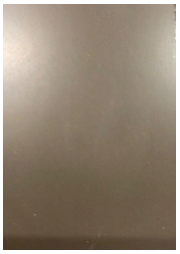
Processing Shape P	Composition X
P_1 steamrolled	X_1 pure
P_2 polished	X_y percentage of main metal component in y %
P_3 casted	

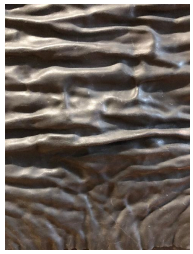
1 Ferrous

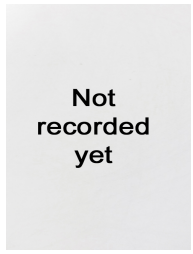


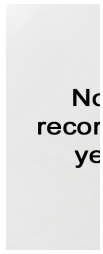
Processing Shape P	Composition X
P_1 steamrolled	X_1 pure
P_2 polished	X_y percentage of main metal component in y %
P_3 irregularly folded	
P_4 perforated (< 5mm)	
P_5 perforated (> 5mm)	
$_6$ rhombic mesh (< 5mm)	
$_7$ rhombic mesh (> 5mm)	
P_8 corrugated	
P_9 square mesh	

2 Non-Ferrous

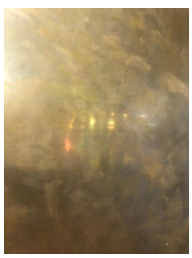
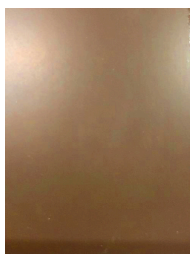








3 Alloys

Processing Shape P	Fineness X (0000, 000, 00, 0, 1,...,5)

P_1 woven

X_{0000} very fine
X_5 coarse

4 Metal Fibers

5 Semiconductors

Not recorded yet

8 Composites

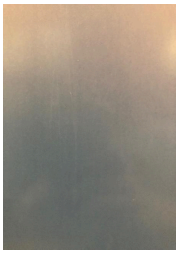
Processing Shape P
P_1 pressed
P_2 covered/mixed
P_3 steamrolled
P_4 woven
P_5 sanded

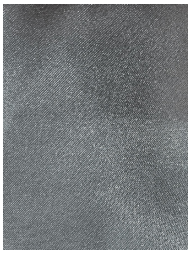
Composition X
X_1 pure
X_y percentage of main component in y %

1 Organic - Polymers

2 Polymer - Stone

3 Polymer - Metal





Processing Shape P	Grit Size X (times 10)
P_1 flat	X_4 P 40
	X_8 P 80
	X_{15} P 150
	X_{24} P 240
	X_{60} P 600

4 Wood - Stone

Not recorded yet

Processing Shape P	Composition
P_1 woven	X_y percentage of component in y %

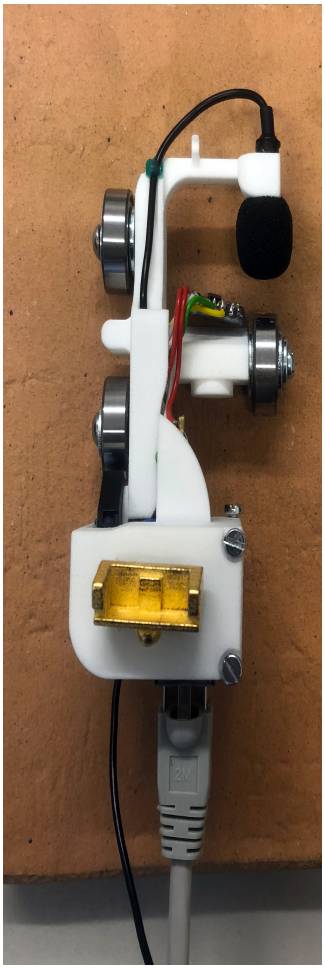
5 Wood - Glass

p

Current Setup - Texplorer 2

he following video shows our Texplorer2 device, the novel recording procedure, parametrization, and display using a novel tactile display device (TCM2).

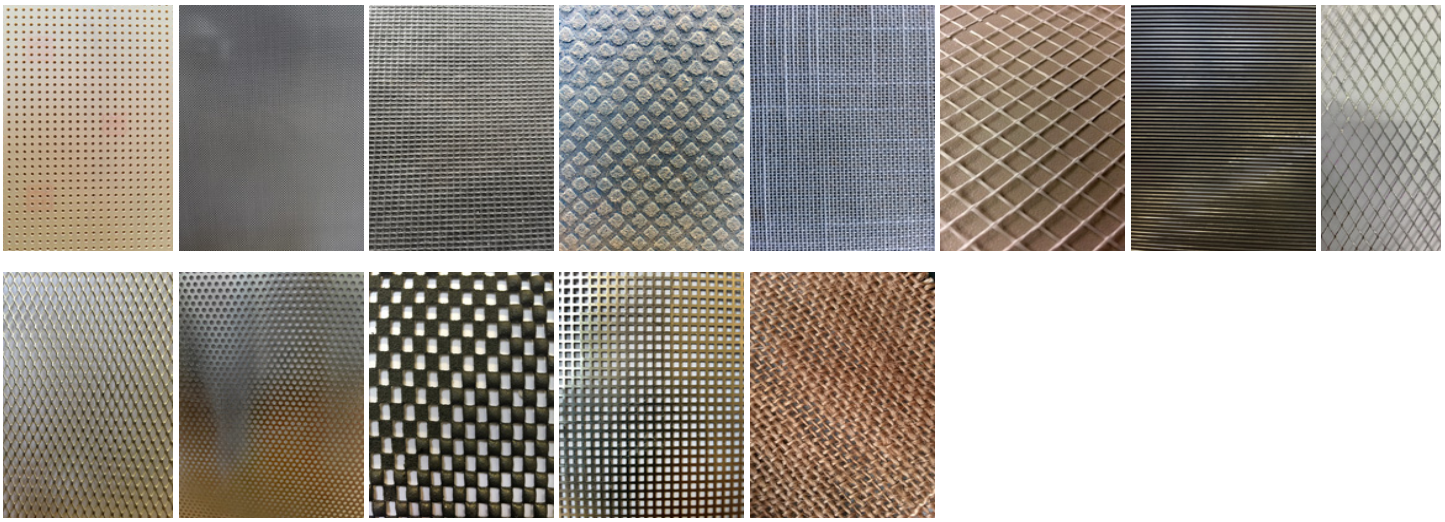
0:00 / 2:51



New Texplorer2 device.

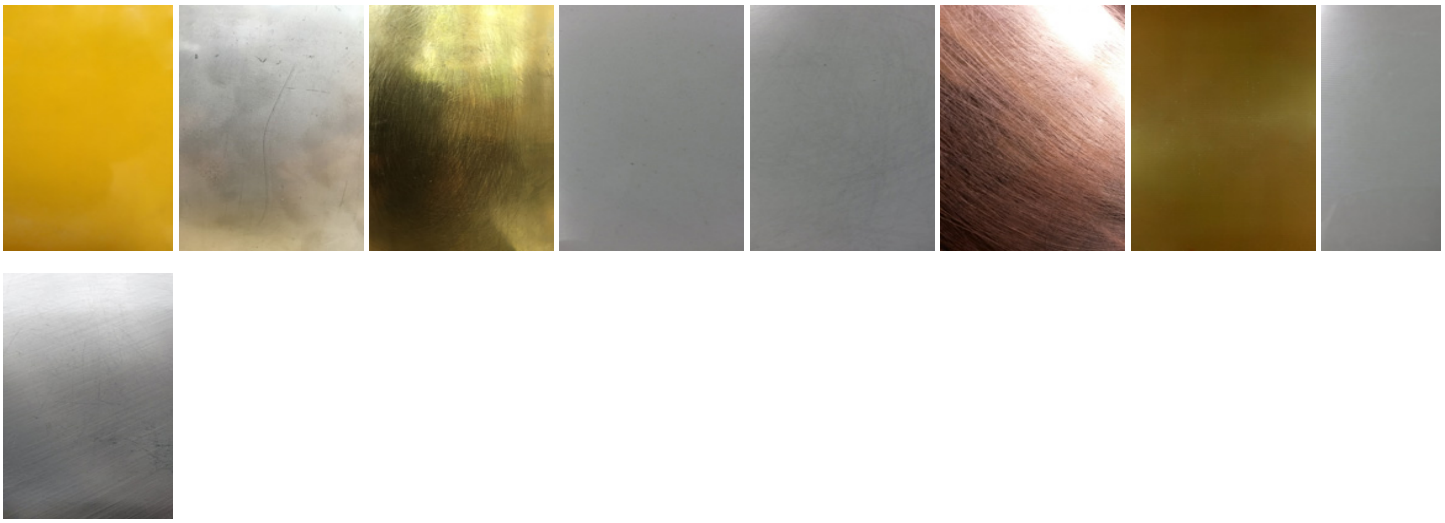
)ld 108 Texture Database

) Meshes



) Stones

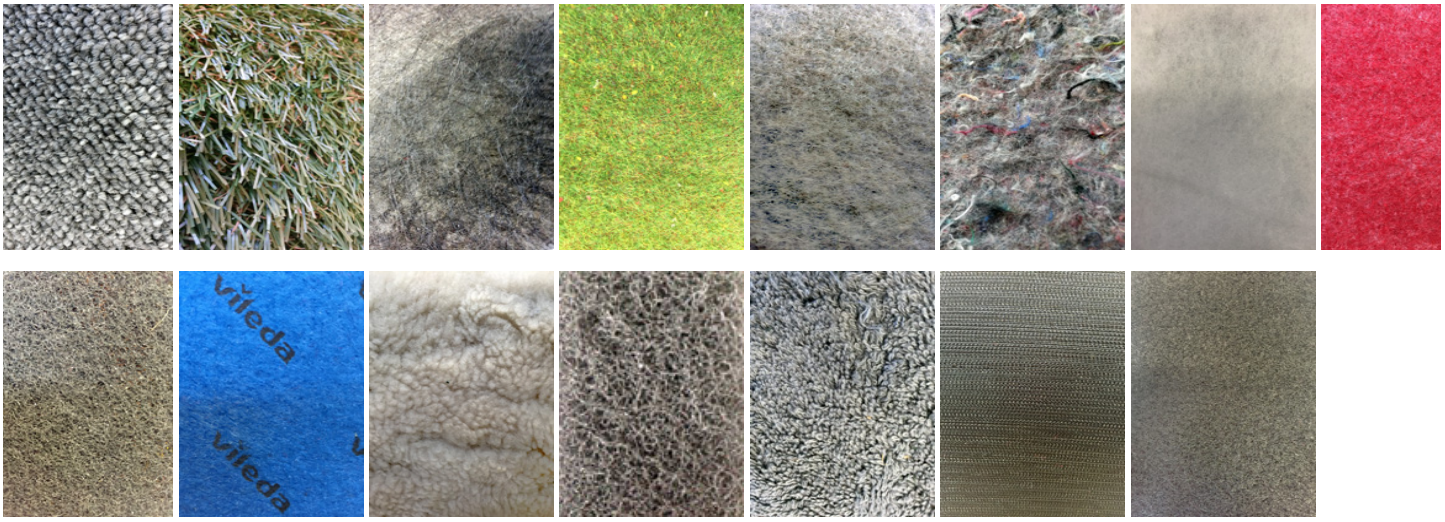
) Blank glossy urfaces



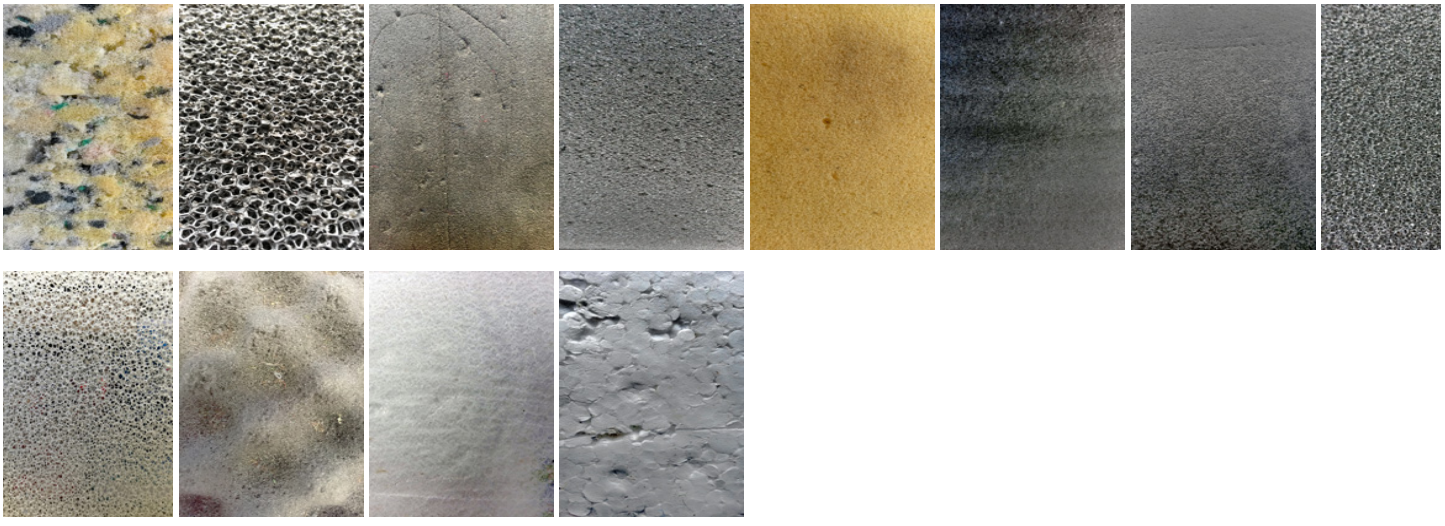
) Wood Types

Rubbers

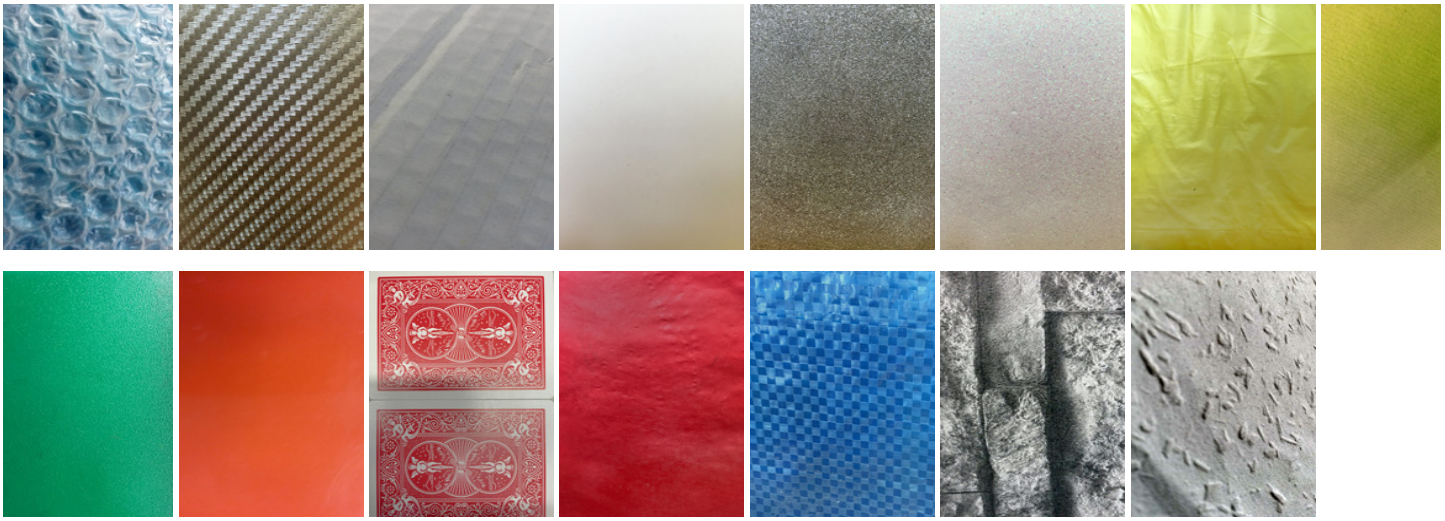
Fibers



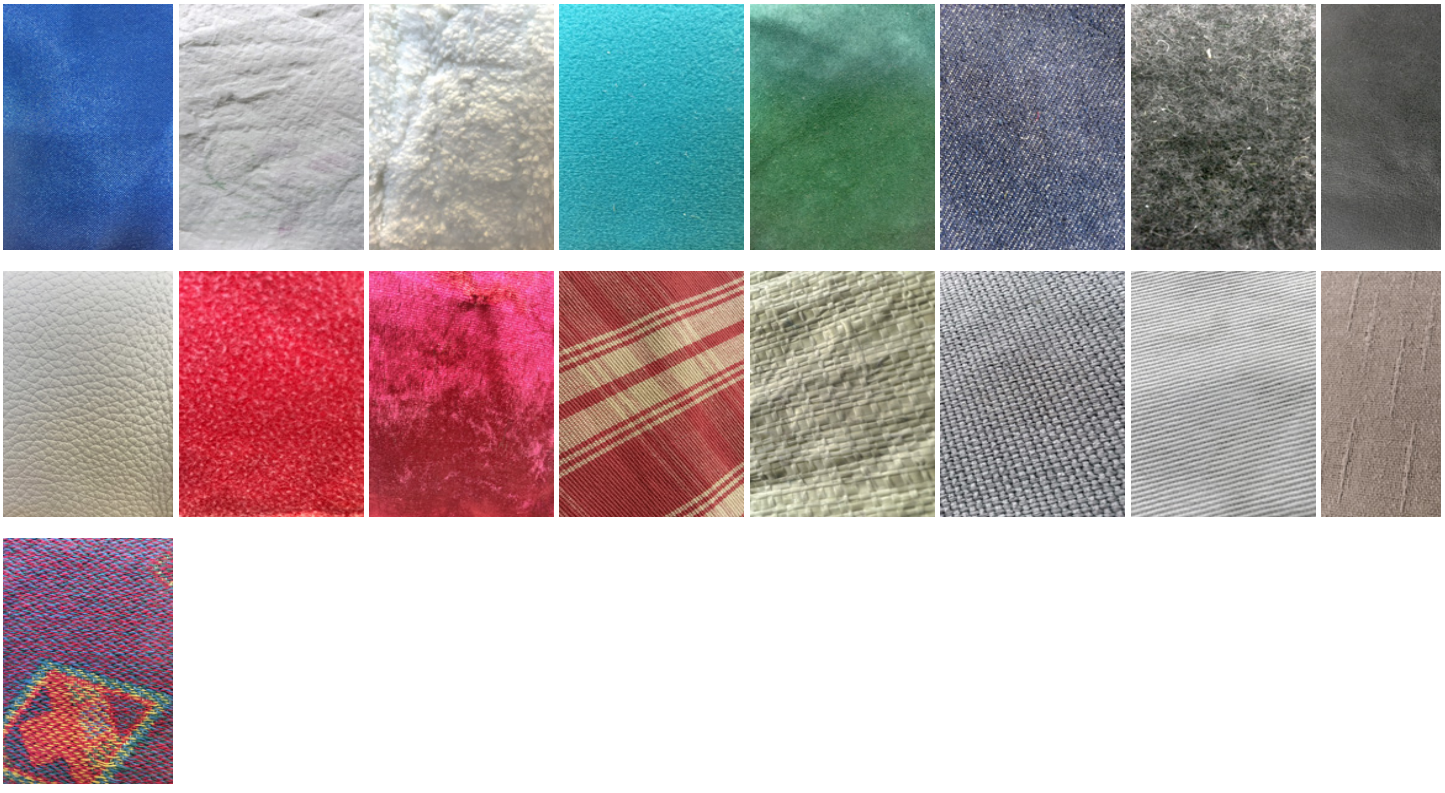
Foams



) Foils and apers



) Textiles and abrics



p

)ld Recording Setups

vo sets of data are recorded for this database.

ontrolled Recordings

rst, the textures are scanned under controlled conditions. Herein, all parameters are kept constant, except the applied force and scan velocity, which are explici ntrolled to allow for a systematic signal analysis. A specialised setup consisting of a rotary texture attachment and a PID-controlled Phantom Omnni is impleme is purpose (compare to Fig. 1) . The inclination angle of the Omni is fixed, while either the applied force or the scan velocity is increased linearly. The main purp is controlled recordings is to test features and their variance against those two scan-time parameters. The following video gives an overview of the robot-contro tup and recording process for the LMT Haptic Texture Database using a PhantomOmni(Sensable) device.

ıe following figures illustrate the hardware setup used for the texture recording. Left: complete setup with a customized rotary plate used to record the c gnals. Center: The two marked joints are fixed to ensure upright posture of the stylus during the recording. Right: close-up of the customized stainless steel toc e accelerometer mounted on the stylus.

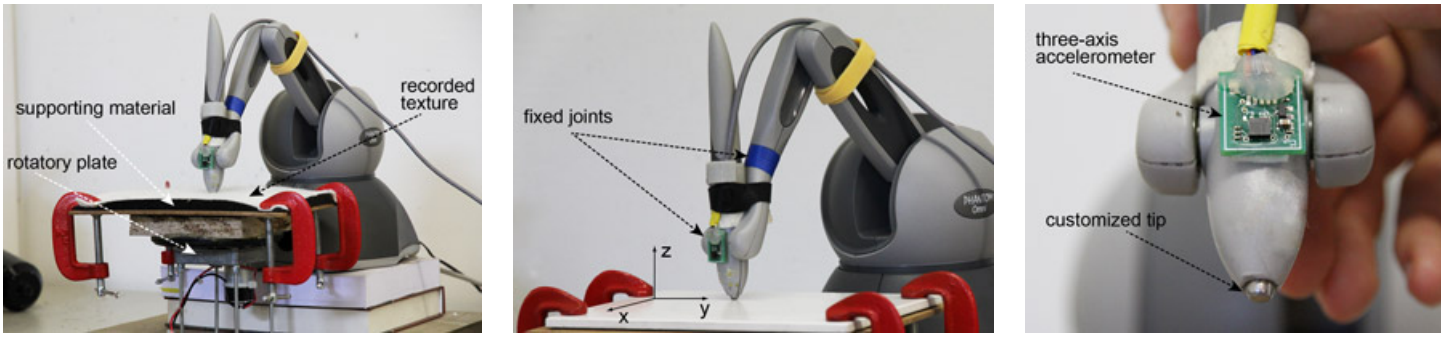
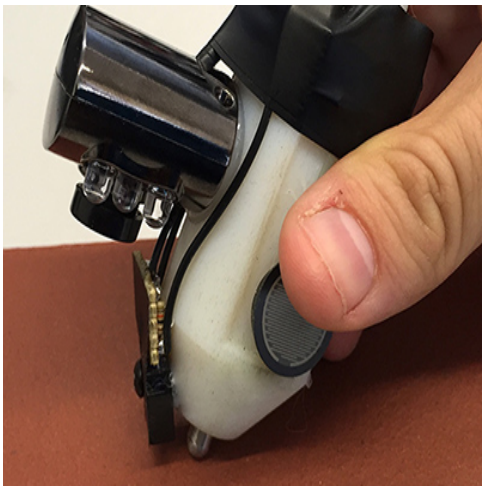


Figure 1 (adopted from [1])

nconstrained Recordings during Freehand Surface Material Exploration

ıe second set of texture signals is recorded under less constrained conditions. It consists of ten free hand recordings for each texture (five linear and five circula ovements). These texture signals, recorded by a single subject following typical human exploratory movements, can serve as a test and training set for texture assification and recognition systems.



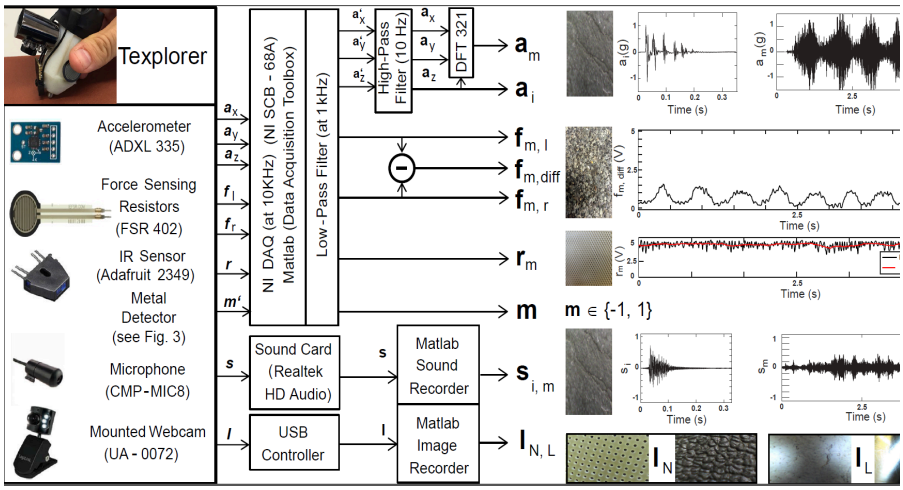


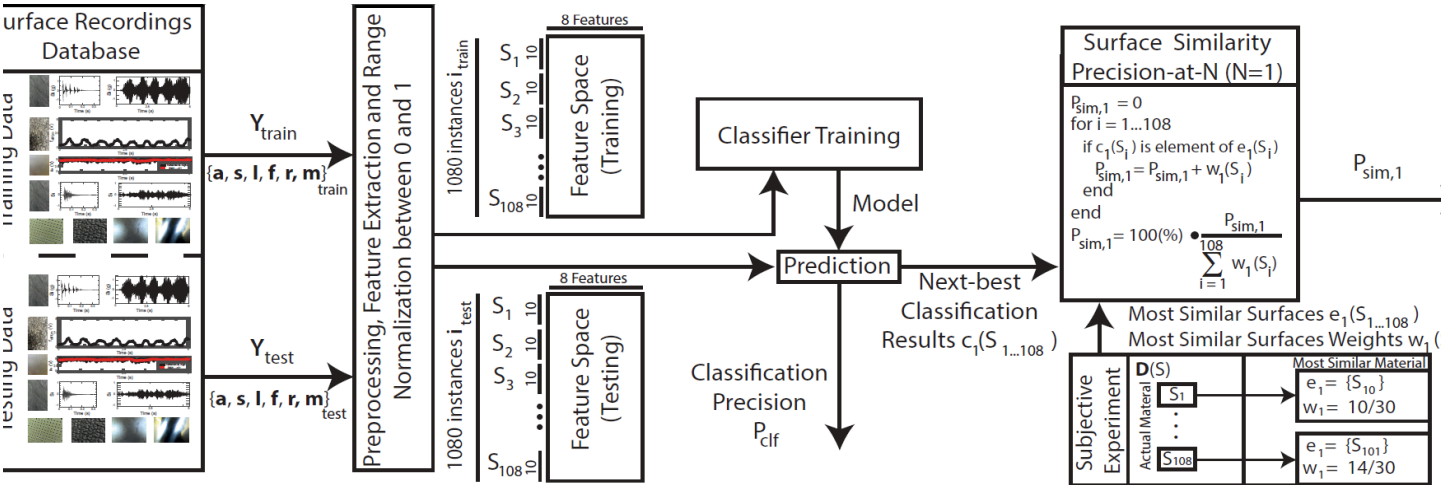
Figure 1 (adopted from [2])

ıe following videos give an overview of the Texplorer device and how a surface material is recorded.

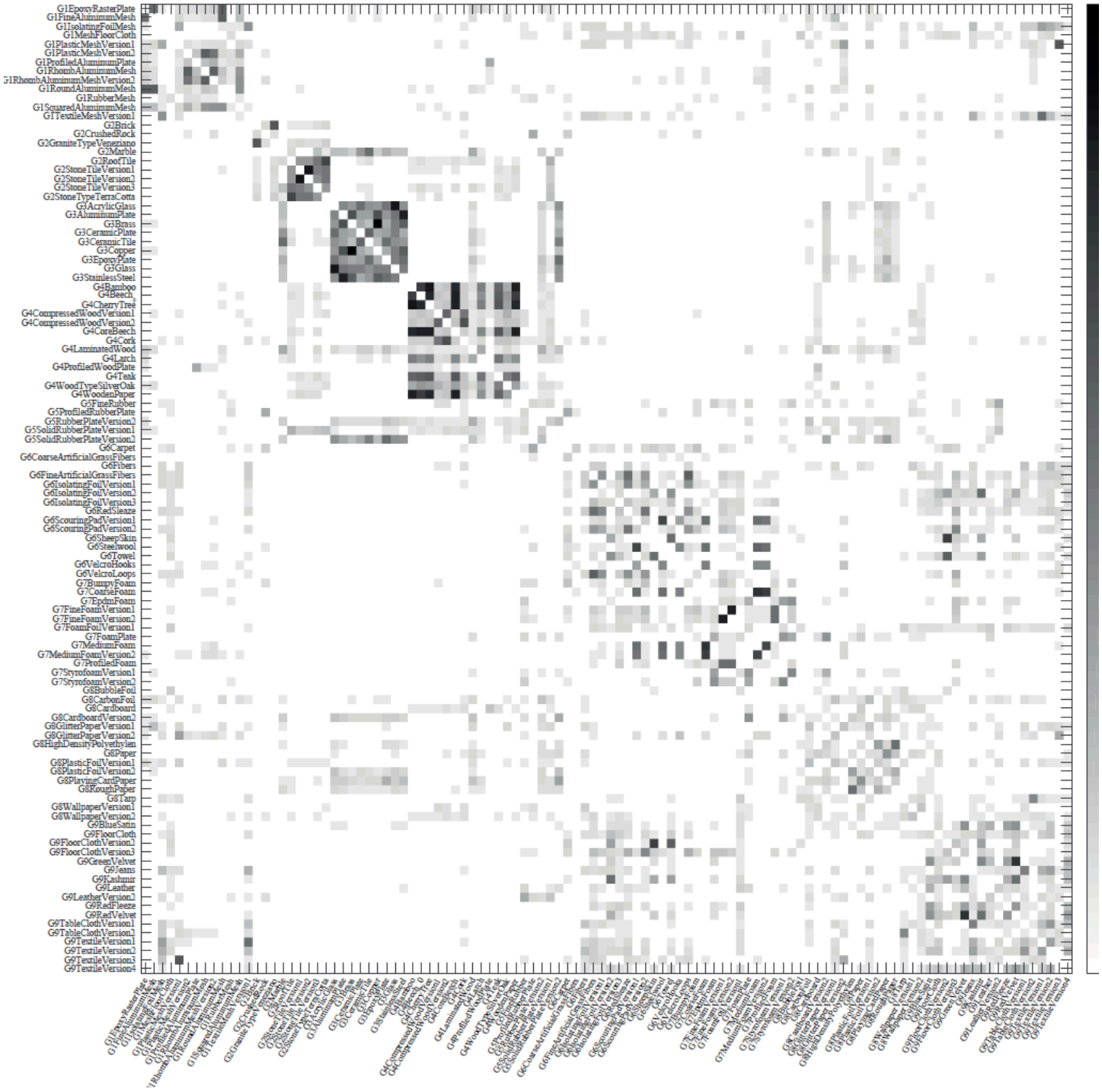
0:00 / 0:05

0:00 / 0:19

ıe following image shows the used retrieval engine for surface material classification and retrieval of the most similar surface materials. Note that we use per otivated features for surface material classifiaction. For perceptual evaluation, we use the outcome of a subjective experiment with 30 subjects to deter erceived similarity between all surface materials.



ıe following image shows the perceptual similarity between all our surface materials.



obotic Texplorer2

ne following video shows our robotic Texplorer 2 setup.

0:00 / 0:29

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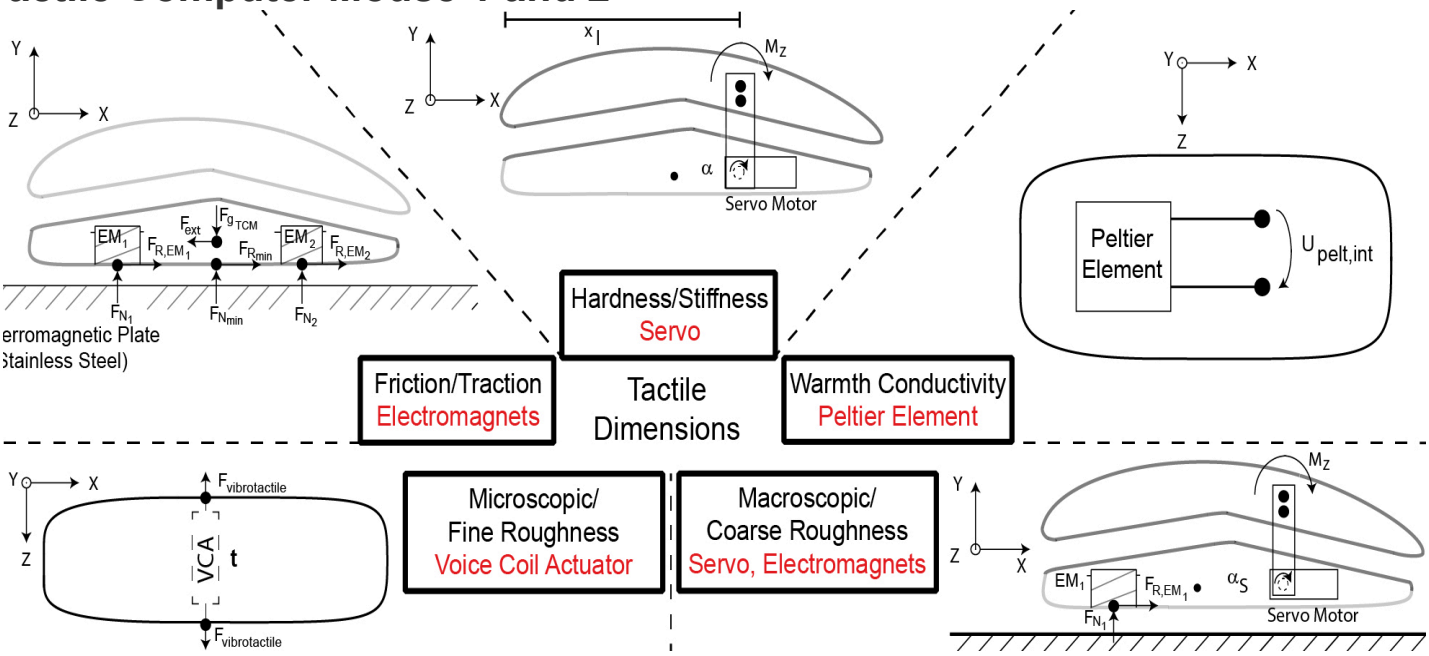
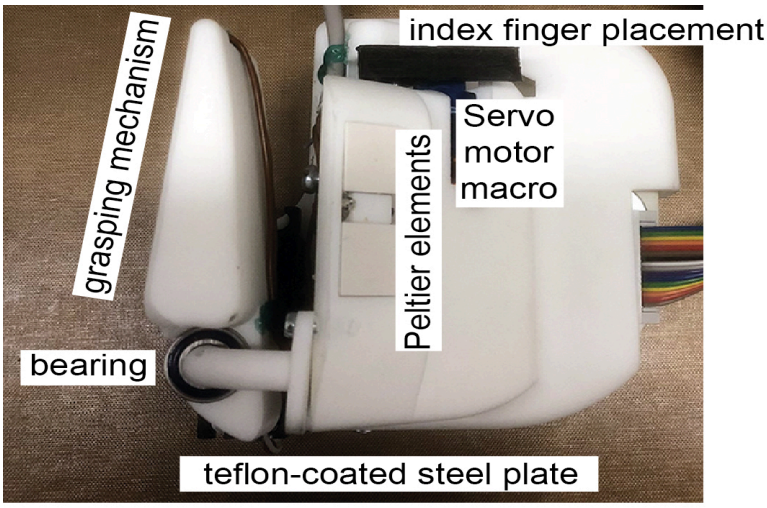
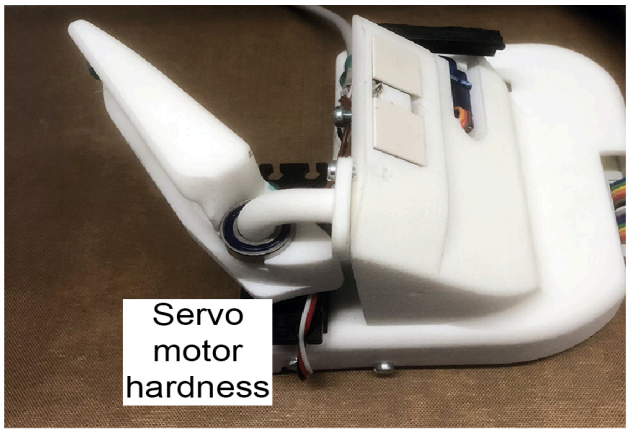
'actile Computer Mouse 1 and 2

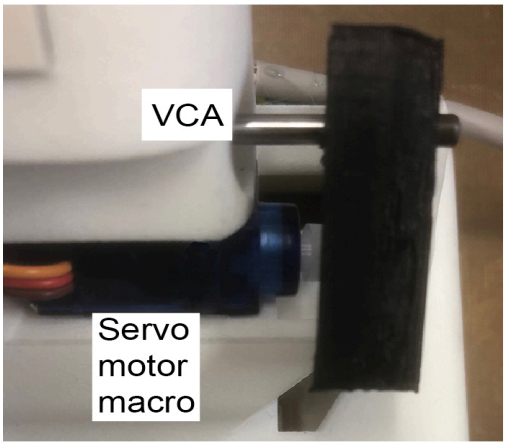


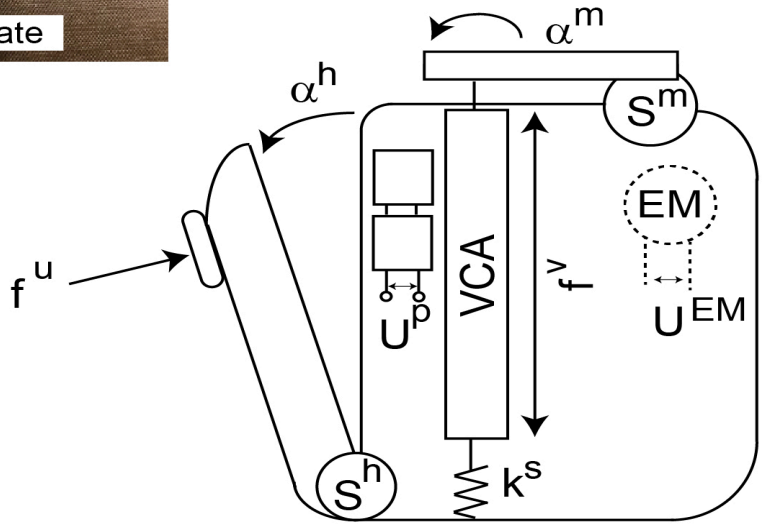
ıe following video shows our Tactile Computer Mouse prototype.

0:00 / 0:28


grasping mechanism
index finger placement
Servo motor macro
Peltier elements
bearing
teflon-coated steel plate


Servo motor hardness


VCA
Servo motor macro


α^m
α^h
S^m
EM
f^u
U^p
VCA
f^v
U^EM
S^h
k^s

p

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ownload

icence

ownload database - 184 materials (from TUM Collection of Building Materials) (zip file, 83484.0 MB)

ownload database description) (txt file, 2 KB)

ownload database (108 surface materials) (zip file, 3190.0 MB)

ownload Similarity Results (zip file, 72.0 KB)

ownload database (69 surface materials) v.1.4 (zip file, 6934.0 MB)

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Chair of Media Technology
Technische Universitaet Muenchen

https://www.lmt.ei.tum.de

Arcisstrasse 21
Stammgelaende, Gebauude 9
80333 Muenchen

Persons involved:

Matti Strese
Clemens Schuwerk
Rahul Chaudhari
Eckehard Steinbach

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